Salary Data Analysis - Age, Education Level and Salary

1. Executive summary

Age and education level are both strongly associated with salary in this dataset of 6,703 employee records. Older employees earn more (Pearson r = 0.728, p < 0.001), and mean salary rises at every education step, from about $36,700 for High School to about $165,700 for PhD (one-way ANOVA F(3, 6694) = 1631, p < 0.001).

Two objectives were tested in R with Pearson correlation, one-way ANOVA and Tukey HSD post-hoc comparisons. Both null hypotheses were rejected. These are associations within one sample, not proof of cause, and the analysis does not yet control for job title, experience or gender.

1.1 Introduction and dataset

Salary is shaped by many personal and professional factors, and employers, analysts and job seekers all want to know which ones matter most. This report examines two of them, age and education level, using the Salary_Data.csv dataset.

The dataset holds 6,703 records with six variables. A few values are missing (1 for Age, 4 for Salary, 2 for Education Level), so each test uses the complete cases for its own variables.

Variable	Type	Role	Summary
Age	Numeric (years)	Predictor	21 to 62, mean 33.6
Gender	Categorical	Descriptive	Male 3,674; Female 3,014; Other 14
Education Level	Categorical	Grouping factor	High School, Bachelor, Master, PhD
Job Title	Categorical	Descriptive	Not used in this analysis
Years of Experience	Numeric (years)	Descriptive	Not used in this analysis
Salary	Numeric (USD)	Target variable	$350 to $250,000, mean $115,327

1.2 Objectives and hypotheses

The study has two objectives, each tested against a null and an alternative hypothesis.

	Objective 1	Objective 2
Aim	Examine the age and salary correlation and its pattern	Test whether salary differs across education levels
H_0	There is no significant correlation between age and salary	Mean salary is the same across all education levels
H_1	There is a significant positive correlation between age and salary	At least one education level has a different mean salary
Test	Pearson correlation	One-way ANOVA with Tukey HSD

1.3 Methodology

All work was done in R and documented in an R Markdown file. The significance level was 0.05 throughout.

Data preparation. Education labels that meant the same thing were merged ("Bachelor's" and "Bachelor's Degree" became Bachelor; "Master's" and "Master's Degree" became Master; "phD" and "PhD" became PhD). Empty education entries were set to missing, and the column was converted to a factor with four levels: High School, Bachelor, Master and PhD.

Objective 1. Pearson's correlation coefficient between Age and Salary was calculated on complete cases, with a significance test (`cor.test`).

Objective 2. A one-way ANOVA (`aov`) compared mean salary across the four education levels. Because ANOVA only says that some difference exists, Tukey's HSD test was then used to compare every pair of levels while controlling the family-wise error rate.

2. Results: age and salary

Age and salary have a strong positive correlation ($r = 0.728$, $p < 0.001$), so the null hypothesis is rejected.

Statistic	Value
Complete cases (n)	6,699

Statistic	Value
Pearson r	0.728
95% confidence interval	0.717 to 0.739
t (df = 6,697)	86.9
p-value	< 0.001
Variance in salary shared with age (r^2)	about 53%

A scatter plot shows the points rising steadily from lower left to upper right. A straight line fitted through them adds about $5,000 to salary for each extra year of age. Above age 50 the points flatten out, with salaries mostly between $150,000 and $200,000, so the straight-line fit is a simplification at the older end.

The result supports age being a meaningful predictor of salary in this sample. Correlation does not show that age itself raises pay: older staff also tend to have more experience and seniority.

2.1 Results: education level and salary

Mean salary differs significantly across education levels ($F(3, 6694) = 1631$, $p < 0.001$), so the null hypothesis is rejected. Education level explains about 42% of the variation in salary (eta squared = 0.42).

Education level	n	Mean salary	Std. dev.	Median
High School	448	$36,707	$22,549	$30,000
Bachelor	3,021	$95,083	$44,092	$80,000
Master	1,860	$130,112	$40,641	$130,000
PhD	1,369	$165,651	$34,340	$170,000

Source	df	Sum of squares	Mean square	F	p-value
Education level	3	7.88×10^{12}	2.63×10^{12}	1631	< 0.001
Residuals	6,694	1.08×10^{13}	1.61×10^{9}		

The Tukey HSD test found that every pair of education levels differs significantly (adjusted $p < 0.001$ for all six). Each step up in education is linked to higher pay, and the largest gap is between High School and PhD.

Comparison	Mean difference	95% CI
Bachelor – High School	$58,376	$53,155 to $63,597
Master – Bachelor	$35,029	$31,990 to $38,069
PhD – Bachelor	$70,569	$67,209 to $73,929
Master – High School	$93,405	$87,978 to $98,833
PhD – High School	$128,945	$123,331 to $134,558
PhD – Master	$35,539	$31,867 to $39,212

3. Discussion and limitations

The findings are consistent: pay rises with age and with each higher education level. Several limits apply to how far they can be pushed.

- **Association, not cause.** Age and education are likely tangled up with experience and job seniority, which this analysis does not separate out.
- **ANOVA assumptions.** Residuals are close to symmetric (skewness 0.34), but Levene's test shows unequal spread across the groups ($p < 0.001$), because High School salaries vary much less than Bachelor salaries. With 6,694 residual degrees of freedom ANOVA is robust to this, and a Kruskal-Wallis test, which does not assume equal variances, gives the same conclusion ($p < 0.001$).
- **Correlation is linear only.** Pearson's r measures straight-line association, and the scatter plot shows salaries levelling off at older ages.
- **One-sided hypothesis.** H_1 for Objective 1 predicted a positive correlation, while `cor.test` runs a two-sided test by default. With $r = 0.728$ the conclusion is the same either way.
- **Data quality.** A handful of records have missing values, and four salaries are under $5,000, which may be entry errors. They are too few (4 of 6,699) to change results of this size, but they would be worth checking at source.

3.1 Conclusion and recommendations

Both null hypotheses were rejected: salary rises strongly with age ($r = 0.728$) and differs significantly across all four education levels ($F = 1631$, $p < 0.001$), with PhD holders earning the most and High School graduates the least.

The results can inform pay-band design, hiring benchmarks and HR planning, as long as they are read as patterns in this sample rather than rules for individuals.

Recommended next steps:

1. Fit a multiple regression of salary on age, education, years of experience, gender and job title, to see which effects survive once the others are held constant.
2. Check whether the education effect differs by gender or job title using an interaction or two-way ANOVA.
3. Review the few extreme or very low salary records before any wider use of the data.